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The principles of dust emission monitoring: Part 1

May 2005, Analytical Instrumentation & Environmental Monitoring

For environmental monitoring purposes it is vital to ensure that the data reported from each emission source is directly comparable and referenced to a known identifiable standard.

The two basic methods for reporting are to identify emissions as mass concentration in mg/m^3 or as a mass flow in gm/sec or kgm/hr . Providing all the necessary information is available, it is possible to convert one standard into the other. Both methods are consistent.

Mass concentration

The most common form of emission reporting is to express the data as a mass concentration. It is important to recognise what is meant by mass concentration.

Dust released to atmosphere from an industrial process is in the form of small particles suspended in a flow of air or exhaust gas. Assuming a homogeneous mixture of dust particles and air, if we were to microscopically examine the emissions we would find that within each cubic metre of air emitted there would be a number of dust particles, and if we knew the particle size distribution and density of those particles we could quickly calculate the mass of dust within that cubic metre of air. Alternatively, we could theoretically filter the cubic metre of air, collect the particles and weigh them, which is precisely what happens in an isokinetic sample test, where a sample of gas is extracted from the duct and the dust particles collected on a filter paper.

The result would be the same: a mass of dust per cubic metre of air, otherwise known as mass concentration.

Suppose that the process producing the dust had constant characteristics and that the rate of mass emission of dust was constant. Suppose then that we were able to double the flow of air that carries the particles by, say, injecting clean air into the duct. While the total mass emission would be constant, we would have diluted the suspension of dust particles in the air by a factor of 2, and the mass concentration would be halved, since the same mass would now be distributed throughout 2 cubic metres instead of one.

Similarly, if instead of increasing the amount of air, we increased the air temperature from 300K to 600K, we would also double the volume occupied by the air, assuming that the pressure remained constant. This too would produce a reduction in the mass concentration of dust by a factor of 2. The same effect occurs if the air pressure reduces. A reduction of 5% in atmospheric pressure will produce a 5% increase in the volume occupied by the air and a 5% decrease in mass concentration.

The measurement of mass concentration is thus dependent on any factor that can change the volumetric characteristics of the carrier gas, be it air or combustion exhaust gases. These factors fall into two groups:

- a) Gas law effects: the effects of temperature and pressure.
- b) Dilution effects: the effects of excess air and water vapour levels.

Data normalisation

It is clear that in order to report dust emissions it is necessary to ensure that the data is presented under a set of standard conditions for describing the volume of the carrier gas. It is standard practice to report the data as a mass per normal cubic metre of dry gas, at a specified level of oxygen.

This means that the actual measured concentration is corrected back to a value consistent with the carrier gas being at 0°C and 101,3 kPa pressure. This takes care of any variation in results due to temperature and pressure.

For measurements of emissions from combustion sources the dilution factor due to ingress of excess air can be determined from a measurement of the oxygen level of the exhaust gas. Since different processes require different levels of excess combustion air for optimum operation, the level of oxygen quoted for the emission normalisation varies from process to process, but must always be stated for the report to be complete and meaningful.

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Exhaust emissions from combustion sources always contain gaseous water, produced from oxidation of the hydrogen within the fuel. The level of water vapour present depends upon the type of fuel being burned. In order to have a unique identifiable standard, it is usual to define the normal gas as a dry gas, thus removing errors in data reporting due to variations in water vapour levels in the emitted gas.

The consequences of all this for dust monitoring from combustion processes can become quite serious. Instead of a relatively simple measurement of dust concentration, in order for the data to be rigorously reported, it becomes necessary to monitor levels of exhaust gas temperature, pressure, oxygen and water vapour. A simple measurement has suddenly become a complex measurement in which the cost of measuring the gas normalisation parameters is greater than the cost of the primary dust measurement. For Schedule A processes, where the normalisation data is already available as part of the gas analysis requirements, this is not a problem. For many Schedule B processes, which are only required to measure dust, the problem becomes severe.

For non-combustion sources, most of which are Schedule B processes, the problem is not merely one of cost, but one of actual feasibility. In a process where the exhaust carrier gas is air, the effects of dilution cannot be measured or detected, since the oxygen level will always be 20,9%. For example, a process that has an allowed mass emission limit of 100 mg/m³, but is emitting 150 mg/m³ of dust, can merely dilute the exhaust gas with air to bring the emitted concentration below 100 mg/m³ and be technically legal, although the total amount of dust being emitted is not reduced by the dilution.

In such situations, specifying an emission limit in mass concentration becomes a worthless and futile exercise. Reported values of dust emission become meaningless, if they cannot be referred to any standard.

Mass flow

The second method of data reporting is to express the emissions in the form of a mass flow in grams per second or kilograms per hour, representing the total mass of dust emitted per unit time, by that process. The measurement is related to mass concentration.

Mass flow = mass concentration x volumetric flow

= mg/m³ x m³/sec

= mg/sec

Note that for this measurement, no normalisation data is required. The value for mass flow is absolute and does not depend in any way on gas temperature, pressure, oxygen or water vapour values, or on any form of dilution of the exhaust gases.

This is illustrated by the earlier example of dust emissions being diluted by a factor of 2 by doubling the airflow. While the mass concentration is halved the volumetric flow is doubled, leaving the value of mass flow unchanged.

Total mass emissions

While a measurement of normalised mass concentration allows a direct comparison of the dust abatement efficiency of different emission sources, it does not help assess the impact on the environment of those sources.

A small stack of 300 mm diameter emitting 500 mg/m³ of dust at 10 m/sec velocity would be ejecting 1,5 gm/sec of dust into the environment. A large power station stack of 6 m diameter emitting only 100 mg/m³ of dust at 10 m/sec would be producing 10 gm/sec of dust.

It is only mass flow that can be related to environmental impact. From measurements of mass flow it is simple to calculate total mass emissions over any period of time, for individual stacks, industrial sites and geographical areas. Emissions reported in mass concentration must be converted to mass flow in order for this information to be obtained, thus requiring the measurement of both mass concentration and gas flow. (To be continued.)

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